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Dynamic models for computer viruses

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ABSTRACT

Computer viruses are an important risk to computational systems endangering either corporations of all sizes or personal computers used for domestic applications. Here, classical epidemiological models for disease propagation are adapted to computer networks and, by using simple systems identification techniques a model called SAIC (Susceptible, Antidotal, Infectious, Contaminated) is developed. Real data about computer viruses are used to validate the model.

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1. Introduction

Globalization and development of communication networks have made computers more and more present in our daily life. Cellular phone, handheld, laptop, mp3 player and many others electronic devices increased human dependence on computers. In this scenario, the large number of existing computer viruses and their high level of destructivity appear as an important risk factor for corporations and individuals.

Computer viruses are small programs developed to damage the computer systems erasing data, stealing information or modifying the normal operation. Their action throughout a network can be studied by using classical epidemiological models for disease propagation (Piqueira et al., 2005; Piqueira and Cesar, 2008).

Based on the Kermack and McKendrick SIR (Susceptible, Infected, Removed) model (Kermack and McKendrick, 1927, 1932, 1933), dynamical and macroscopic models for computer viruses propagation were proposed, providing estimations for temporal evolutions of infected populations depending on

network parameters considering topological aspects of the network (Kephart et al., 1993; Mishra and Saini, 2007).

This kind of approach was successfully applied to e-mail propagation schemes (Newman et al., 2002) and modifications of SIR (Susceptible, Infected, Removed) models generated guides for infection prevention by using the concept of epidemiological threshold (Piqueira et al., 2005; Mishra and Saini, 2007; Draief et al., 2008).

In previous works of the present group, SAIR (Susceptible, Antidotal, Infected, Removed) (Piqueira et al., 2005) and SAI (Susceptible, Antidotal, Infected) (Araujo, 2004) models were proposed but good fitness to real data was not obtained. Consequently, SAIC model was tried with a satisfactory replication for the real data about computer viruses dynamics.

This paper starts with brief descriptions of the models followed by an explanation about how the model parameters were identified. Comparisons between model outputs and real data are presented showing that the model can be considered adequate to describe the spreading evolution of a computer virus.

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Fig. 1 – SIR model.

2. Compartmental models

There is a vast catalog of compartmental models indicated for epidemiology (Murray, 2002) and their origin is Kermack and McKendrick SIR model (Kermack and McKendrick, 1927, 1932, 1933), as shown in Fig. 1.

This model is composed of three populations: Susceptible – all individuals in this group can acquire the infection when in contact with an infected individual; Infected – all individuals in this group carry and propagate the infection; Removed – the individuals in this group died or were removed from the population.

Dynamic equations for the populations S, I and R are:

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI \\ \frac{dI}{dt} &= \alpha SI - \beta I \\ \frac{dR}{dt} &= \beta I\end{aligned}\quad (1)$$

In this model, the total population $S + I + R$ is considered to be constant and the rate of interactions between susceptible and infected, the infection rate, is represented by constant α and the rate of the removing process, by β .

Based on this model, adding an antidotal (A) compartment and considering an influx rate N , representing the incorporation of new computers to the network, the SAIR model for computer viruses propagation was proposed, as shown in Fig. 2, with the following dynamical equations:

$$\begin{aligned}\frac{dS}{dt} &= N - \alpha SA - \beta_{SI}SI + \sigma_{IS}I + \sigma_{RS}R \\ \frac{dI}{dt} &= \beta_{SI}SI + \beta_{AI}AI - \sigma_{IS}I - \delta I \\ \frac{dR}{dt} &= \delta I - \sigma_{RS}R \\ \frac{dA}{dt} &= \alpha SA - \beta_{AI}AI\end{aligned}\quad (2)$$

The parameters of the model are defined as follows:

- N : influx rate, representing the incorporation of new computers to the network;

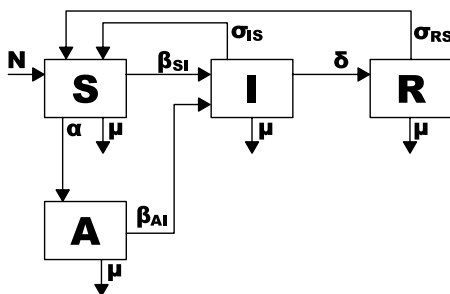


Fig. 2 – SAIR model.

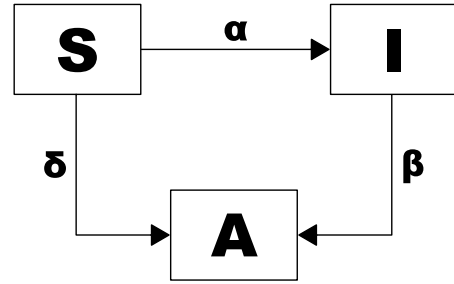


Fig. 3 – SAI model.

- μ : mortality rate not due to the virus;
- β_{SI} : infection rate of susceptible computers;
- β_{AI} : infection rate of antidotal computers due to the onset of new virus;
- δ : removing rate of infected computers;
- σ_{IS} : recovering rate of infected computers;
- σ_{RS} : recovering rate of removed computers, with an operator intervention;
- α : conversion of susceptible computers into antidotal ones, occurring when susceptible computers establish effective communication with antidotal ones and the antidotal installs the anti-virus in the susceptible ones.

SAIR model was analyzed and basal reproduction rate was calculated for short-term behavior, i.e., considering $N = 0$ and $\mu = 0$ (Piqueira et al., 2005).

When one tries to adapt the SAIR model to real data, the R compartment needs to be unconsidered, generating the SAI model (Araujo, 2004), shown in Fig. 3, with the following dynamical equations:

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI - \delta SA \\ \frac{dI}{dt} &= \alpha SI - \beta IA \\ \frac{dA}{dt} &= \delta SA + \beta IA\end{aligned}\quad (3)$$

As reported by Araujo (Araujo, 2004), SAI model presents a better fitting to real data but it can be improved including a new compartment representing infective machines that do not express symptoms. This new compartment will be called

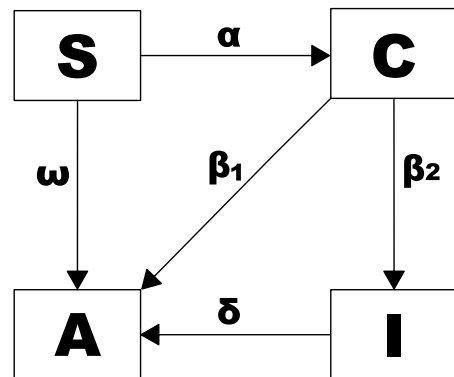


Fig. 4 – SAIC model.

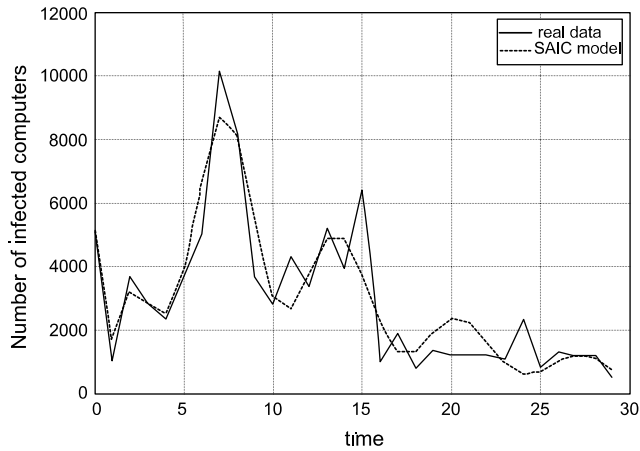


Fig. 5 – worm_netsky_p infection.

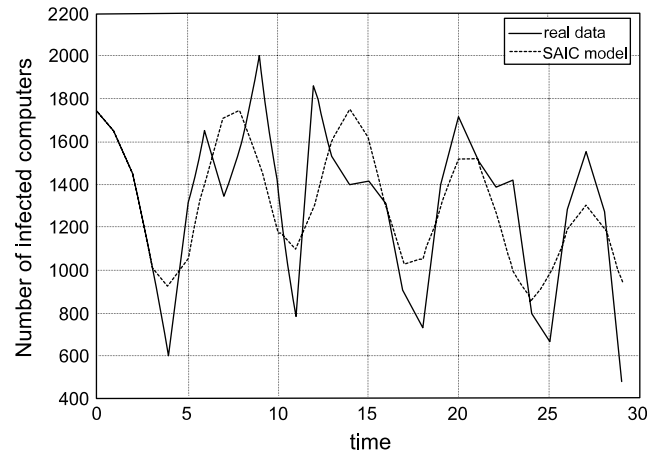


Fig. 7 – troj_swizzor_dq infection.

Contaminated (C) and the SAIC model is represented in Fig. 4, with the following dynamical equations:

$$\begin{aligned} \frac{dS}{dt} &= -\alpha S \frac{\ln I}{\ln A} - \omega S \ln A \\ \frac{dC}{dt} &= \alpha S \frac{\ln I}{\ln A} - \beta_1 C \ln A - \beta_2 C \\ \frac{dI}{dt} &= \beta_2 C - \delta I \ln A \\ \frac{dA}{dt} &= \omega S \ln A + \beta_1 C \ln A + \delta I \ln A \end{aligned} \quad (4)$$

Model parameters are: contamination rate, α , immunization rate, β_1 , infection rate, β_2 , disinfection rate, δ , and immunization rate, ω . Logarithms of the populations were introduced, in order to obtain a good fitting to real data.

In the next section, we explain how the SAIC model was adapted to the data related to the infections by worm_netsky, troj_swizzor_dq and spyw_dashbar_300, showing comparisons between the dynamics of the model and the real infections.

2.1. Fitting the model to real data

In order to adjust the model parameters to real data, the MATLAB function “fminunc” (Lynch, 2004) was used, starting

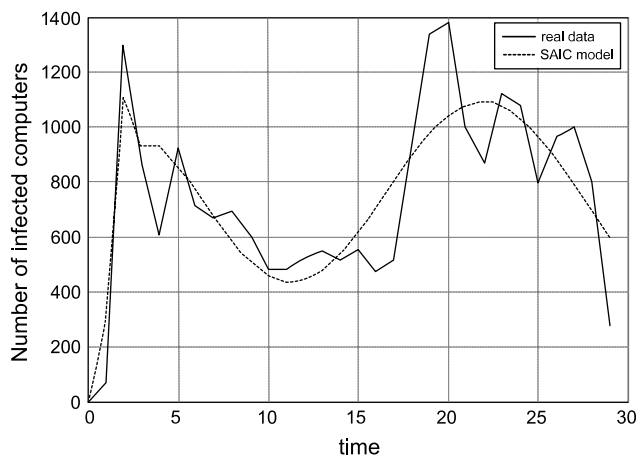


Fig. 6 – pyw_dashbar_300 infection.

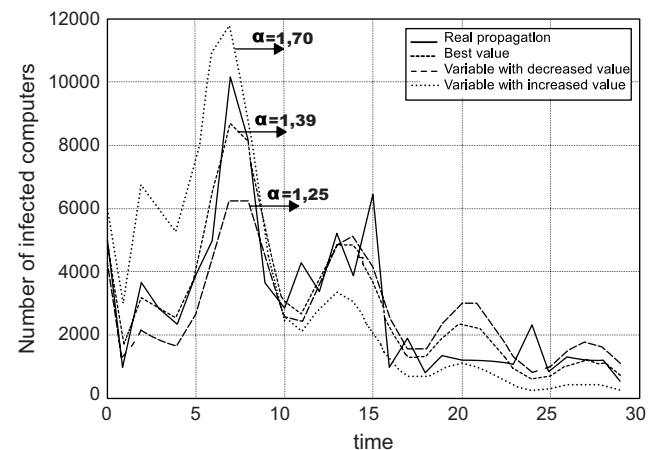
with an initial guess for the parameters and minimizing the mean square error between the infected population given by the model and the experimental data reported in the INTERNET site www.wildlist.org.

Fig. 5 shows the results for the computers infected by worm_netsky_p; Fig. 6, for computers infected by spyw_dashbar_300; and Fig. 7, for computers infected by troj_swizzor_dq. In the three figures, the time scale is in days.

As the results above show, the SAIC model has a satisfactory adherence to real viruses dynamics. A simple parameter estimation procedure is enough to obtain a set of parameters that fits the data.

2.2. Model robustness

The robustness of the model can be evaluated considering small variations in the parameters around the optimal value obtained in the identification process. The worm_netsky_p infection dynamic was simulated varying the contamination rate, α , as shown in Fig. 8, with the time scale in days.

Fig. 8 – Model robustness under contamination rate variations (α).

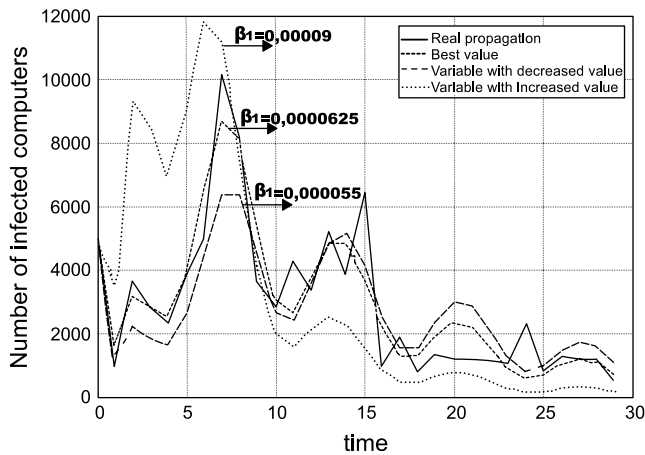


Fig. 9 – Model robustness under immunization rate variations (β_1).

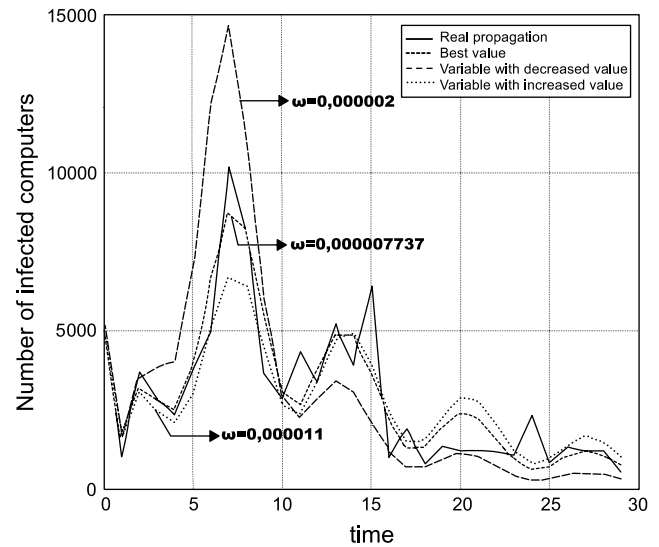


Fig. 12 – Model robustness under immunization rate variations (ω).

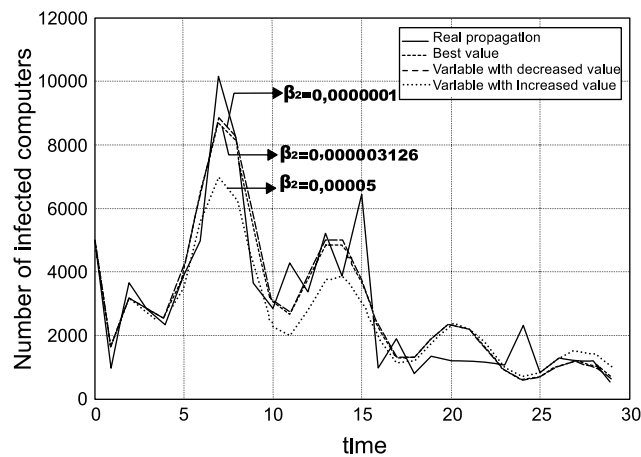


Fig. 10 – Model robustness under contamination rate variations (β_2).

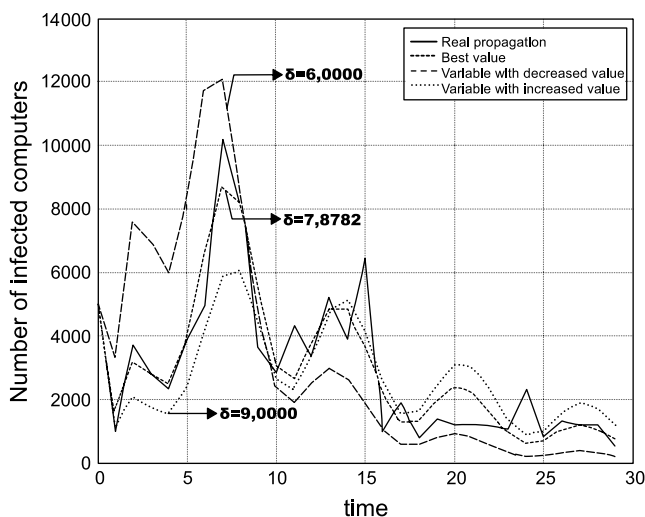


Fig. 11 – Model robustness under disinfection rate variations (δ).

By using the same values around the optimal one and with the time scale in days, results are shown in Fig. 9 for the immunization rate, β_1 ; Fig. 10, for the infection rate, β_2 ; Fig. 11, for the disinfection rate, δ ; and Fig. 12 for the immunization rate ω . The observation of these figures allows concluding that the SAIC model is robust under parameter variation.

3. Conclusion

Inspired on disease propagation models, a compartmental model was developed for the spread of viruses in computer networks. The SAIC model seems to be adherent to the real data in a satisfactory way and, besides, is robust under parameter variations.

As the SAIC model has simple implementation, its use might help in estimating the dynamic behavior of viruses in real systems. Parameter influence is easy to obtain providing tools to establish procedures, in order to attenuate virus damages.

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